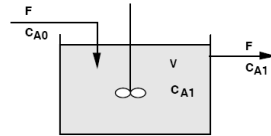
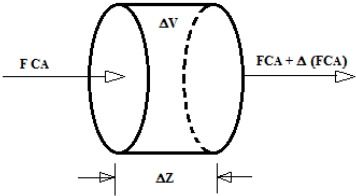


DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE Winter Examination – 2022 Course: B. Tech. Branch : Semester :VII Subject Code & Name: BTCHC705 D & Modeling and Simulation in Chemical Engineering Max Marks: 60 Date: Duration: 3 Hr.			
Instructions to the Students: 1. All the questions are compulsory. 2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question. 3. Use of non-programmable scientific calculators is allowed. 4. Assume suitable data wherever necessary and mention it clearly.			
		(Level/CO)	Marks
Q. 1	Solve Any Two of the following.		12
A)	Explain the general modeling procedure with diagram.	CO1	6
B)	What is the difference between the lumped parameter system & distributed parameter system? Explain with examples.	CO1	6
C)	What are applications of process modeling?	CO1	6
Q.2	Solve Any Two of the following.		12
A)	The penultimate step in cappuccino making is the steaming of milk before the hot milk is poured over the espresso. Consider this process of steaming the milk. Steam is injected from a nozzle immersed in the container of cold milk as shown in the following figure. (i) Comment of whether the steaming process can be considered to be a <i>lumped</i> or <i>distributed</i> parameter system. Provide reason to support your answer. (ii) Comment on whether the system can be considered as steady-state or transient with respect to each of relevant balance equations – (a) Overall Energy Balance (b) Overall Material Balance and 	CO1	6
B)	A plant discharges an aqueous effluent at a volumetric flow rate F. Periodically, the effluent is contaminated by an unstable noxious waste, which is known to decompose at a rate proportional to its concentration. The effluent must be diverted to a holding tank, of volume V, prior to final discharge, as in Fig. Develop a mathematical model for the situation. 	CO1	6
C)	The complex chemical reaction shown below is carried out in an isothermal constant volume batch reactor. Develop model equations and draw information flow diagram.	CO2	6

Q. 3	Solve Any Two of the following.		12
A)	3 CSTR are connected in series of reaction $A \rightarrow B$ is taking place in each reactor. Each reactor is maintained at different temperature and rate constant in each reactor is k_1, k_2, k_3. Volume of reactor is V_1, V_2 and V_3 respectively. A reactant A is fed to first reactor at a flow rate of F_0 at concentration of C_{A0}. Derive mathematical model for the system.	CO2	6
B)	In a bioreactor we assume that the reactor is perfectly mixed and that the volume is constant. X_1=biomass concentration, X_2= substrate concentration, r_1= rate of cell generation, r_2= rate of substrate consumption, F = volumetric flowrate. Write a material balances on the biomass (cells) and the substrate (feed source for the cells) to describe the behavior of the system.	CO2	6
C)	Consider the tank below. Hot water is used to heat the contents of a reactor. It is supplied to the jacket at a temperature of T_s. Heat is transferred at a rate of Q_{in} (J/sec = Watts) from the jacket to the reactor contents. This input causes a change in the reactor temperature, T. The liquid in the reactor has a mass, m (kg), and a specific heat capacity of C_p (kJ/kgK). Heat is removed from the reactor at a rate of Q_{out} (W). Develop a mathematical model.	CO3	6
Q.4	Solve Any Two of the following.		12
A)	Develop a dynamic model for binary system using continuous distillation column.	CO3	6
B)	Develop a model for single stage Extraction process.	CO2	6
C)	Develop a model for stage wise absorption column.	CO3	6
Q. 5	Solve Any Two of the following.		12
A)	Develop a model for Plug flow tubular reactor with assumptions.	CO4	6

			
B)	What are the differences between modular approach and equation oriented approach?	CO4	6
C)	Explain the types of process simulation problems.	CO4	6
	*** End ***		

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